

D1. Process For Generating & Comparing Possible Solution

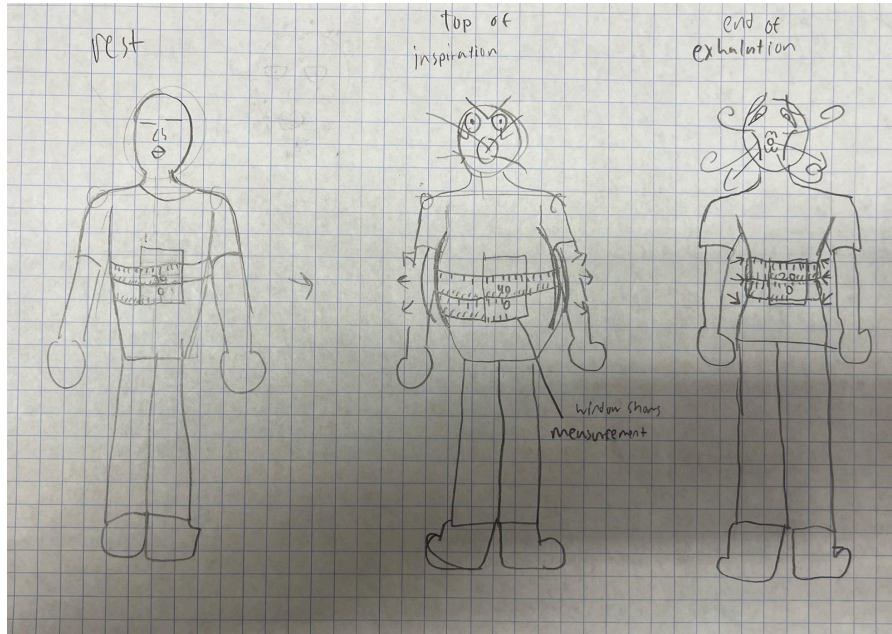
Morphological Chart

Function	Design Idea 1	Design Idea 2	Design Idea 3	Design Idea 4	Design Idea 5	Design Idea 6	Design Idea 7
Measure Upper, Middle, Lower Chest Expansion	Tape measure with window and arrow	Arduino Program with IMU sensors	Yoyo method (Spring Loaded Potentiometer with flywheel on top)	Elastic Belt with Stretch Sensor w/ tape measure	Elastic belt and string encoder	arduino digital tape measure https://youtu.be/tTV52DihPTc?si=loyRII9CSUjvTqI_ Digital tape measure: This works by using a rotational counter to count the amount of rotations as the tape measure lengthens and compressed and converts into a length measurement (circumference*rotations) https://youtu.be/OPsZONVKJq0?si=GQd-ZuOakjyT_vT5	Arduino sound wave sensor https://youtu.be/sZjN_KuHzRw?si=wkUm cP42 oW1t0 Dc
Measure Breathing Rate	Arduino measures the successive stretches as the patient breathes, converting	Arduino computes a graph measuring the oscillation between the max and min	Arduino measures how many times the potentiometer senses a turn in the	Pitch counter hooked up to tape measure with arrow. (turn pull	Arduino Breathing monitor (for breathing rater)  Make a...	Fetalplus	

	into breaths per min	circumferences	positive direction and converts that into breaths per min	into push force)			
Hold Shape and Position During Each Breath	Patch openings throughout. (maybe close through velcro)	Sewn directly into the onesie (can't be washed).	Woven into the fabric	Onesie that houses all measurement components	Adjustable vest	belt	
Measure Heart Rate	dry-electrode ECG monitors	Optical Sensor	Photoplethysmography - https://pubmed.ncbi.nlm.nih.gov/17322588/	electrocardiogram (ECG) - https://my.clevelandclinic.org/health/diagnostics/16953-electrocardiogram-ekg	Does not measure heart rate	Cardiac Event Monitor	
Able to withstand continued use	Machine washable (parts and sensors able to be taken out)	Modular	Stretchable (can be used for multiple body types)	Replacement parts available			
Data Display	Do not display data (record by hand)	Calculator program that can give the difference	Connect to an arduino program sending results to a digital display	Create an application that stores	Use esp32 to wirelessly display to a digital system		

Conceptual Designs

Design idea 1: Tape Measure With Window



Tape measure is wrapped about the inside of the onesie. The measure is attached to an elastic band so it can stretch and compress according to the chest circumference. There will be a window to show the measurement of the tape measure

This design contains tape measure/s that are wrapped around the onesie. One end of the tape measure is static and the other can move freely but still have some sort of light elastic to help it return to its original position during an exhale. The window then shows the measurement of how much the stomach circumference had changed.

Strengths	Weaknesses
• Easy to manufacture • Low cost and amount of materials • Intuitive • Able to be added onto in the future. (load cell to measure breathing rate. The load cell could be connected to the elastic band and every inhale that the load cell senses, it counts it and creates a breaths per minute)	• Potentially inaccurate • No digital data sent to an application. Old methods of recording the data. The same way they collect data currently. (This could cause the former inaccuracies)

Design idea 2: Digital String Encoder Arduino

This design focuses on using encoder technology to record the chest circumferences and store the measurements into an arduino to provide accurate measurements and a method to store data digitally. This device can convert changes in electrical currents to a length measurement. We can then use the measurements to compute the absolute value difference in chest circumference at the top of inspiration and end of exhalation.

Digital Encoder:



This device reports a linear value dependent on the number of rotations

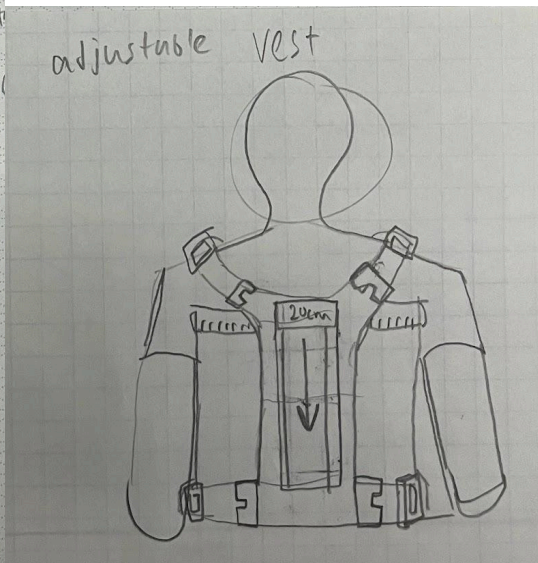
We can use the encoder to record the measurements of the circumference. The encoder reports a value but we must use code to convert this value into a measurement we can use for chest circumference. [Code](#)

Strengths	Weaknesses
• Allows for quick storing of measurements into arduino • Digitally shows the measurements and can compute differences between top of inspiration and end of exhalation	• The arduino must be held in a separate compartment on the onesie which can be a hindrance in the comfortability in the suit • Requires an arduino and experience using arduino in order to transfer the data into desired device • Difficult to manufacture

Design 3 - Adjustable Vest

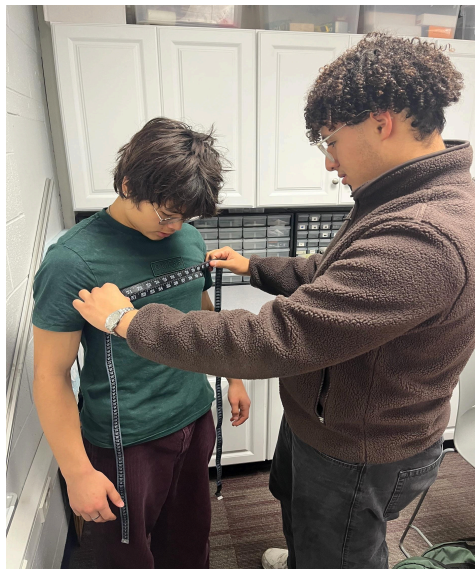
This conceptual design focuses on using a tape measure to measure the chest circumference of the patient. This mirrors the past method Dr. Broseman has used but improves upon it by

making it more accessible as it doesn't require the user to constantly hold the two ends of the tape measure. The design uses a vest mechanism to hold the tape measures. Straps are located on the shoulders and on the waste in order to secure the vest in place. This also allows for the vest to fit a high range of body types. The tape measure is fitted in a slider with adjustable rails in order to measure chest circumference across different sections of the chest. This design doesn't incorporate a heart rate monitor since it is considered a low priority requirement. An arduino will be used to measure breathing rate and, potentially, heart rate.



Strengths	Weaknesses
- This design is accessible because it is very intuitive and easy to use; the user must set the tape measure to the appropriate high then read the measurement to calculate chest circumference - Easy to put together and prototype	- The data on chest circumference must be manually recorded. This can make it time consuming when doing intensive research. - The parameters such as difference in the circumferences at the top of inspiration and end of exhalation

since the highest priority design requirement (chest circumference doesn't require complex coding or electrical components) • We could include a heart rate monitor. It's simple to implement and won't require complex components.	aren't reported and must be interpreted manually.
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The images to the left and on the bottom roughly depict how the tape measure will be used to measure chest circumference

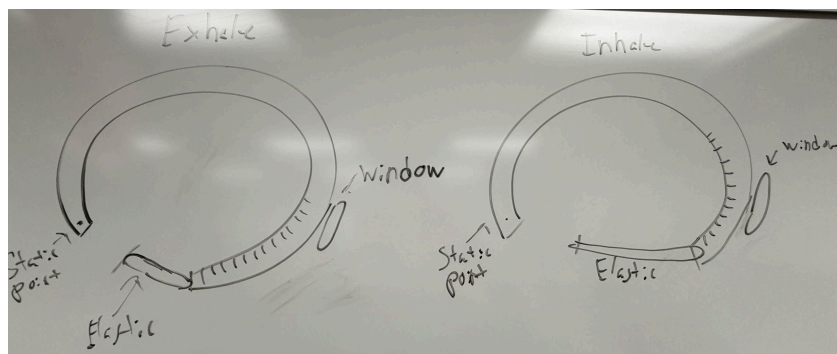
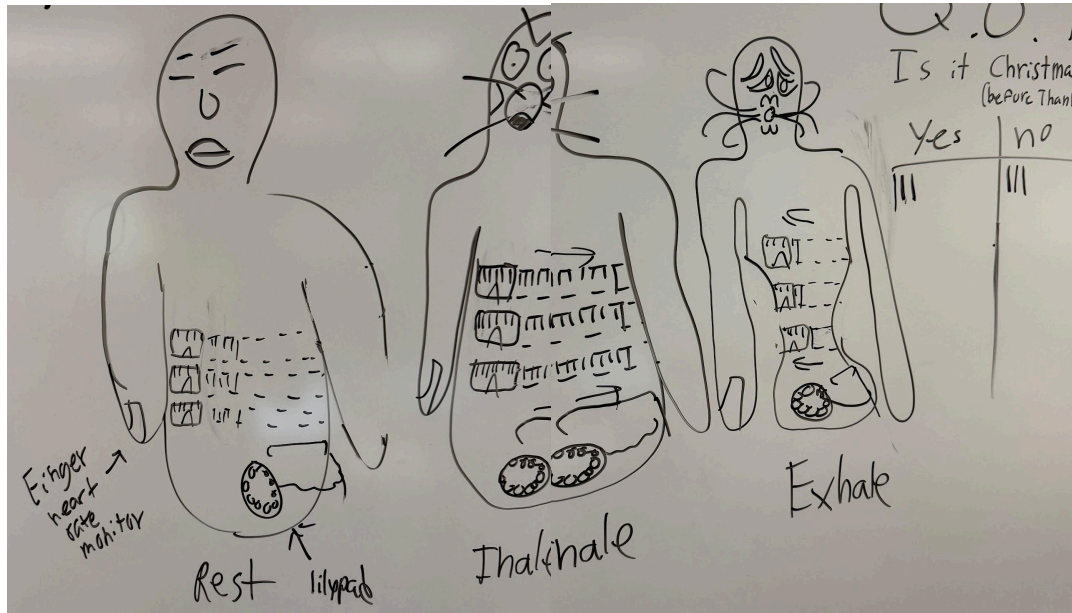


The tap measure will lengthen and compress as the chest expands and contracts. The tape measure can report the chest circumference and

the difference between the circumference at the top of inspiration and end of exhalation.



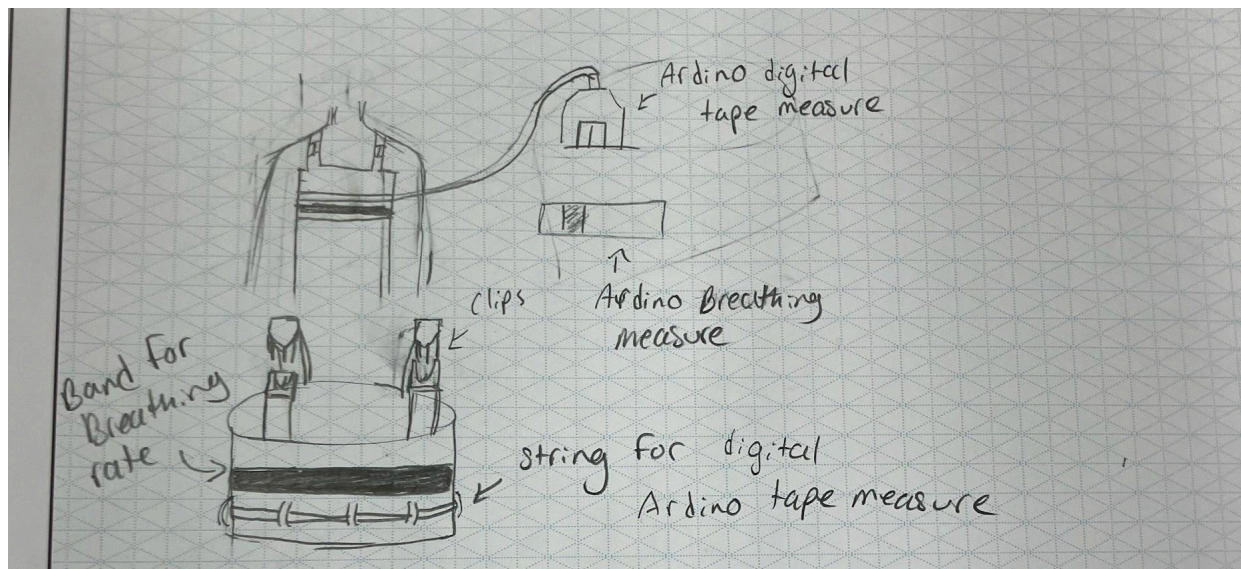
Design 4 - Triple Tape Onesie



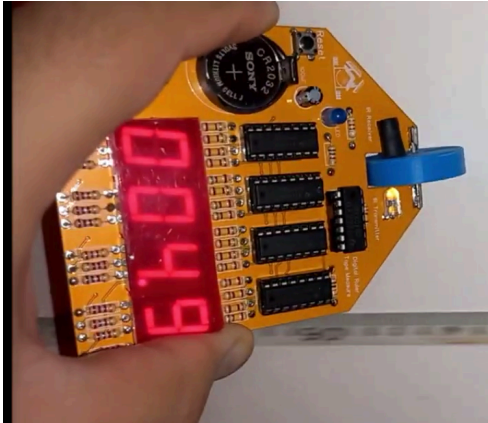
This design is focused on simplicity and convenience. The three separate tape measures are all measuring simultaneously, eliminating the need to adjust the measurement between sessions. This also opens up the opportunity to compare circumferences between the measurements (Middle vs Upper) (Upper vs Lower) (etc.) at the same time. The design also includes a tensioned Stretch Sensor to measure the breathing rate to avoid old-fashioned counting with a timer. Having this arduino sensor allows the design to still have arduino parts, but avoids complex programming that would be needed if a sensor was used for measuring circumference itself. This simple sensor as well as an arduino that can be mounted onto the onesie allows for removal and cleaning if needed. The measurement parts should not be affected when wet if the elastic is made up of correct material. Measuring heart rate is a low priority but I still incorporated it into my design. It would be a store bought finger sensor that could be adapted to different finger sizes.

Strengths	Weaknesses
• Simple measurement process • able to compare different chest locations to each other at the same time. • No complicated arduino wiring and programming. • Electrical parts able to be removed for cleaning • Heart rate incorporation Kid-s-Fingertip-Pulse-Oximeter	• Still having to measure by reading and writing down instead of the numbers appearing in front of you. • Could be bulky having three separate measurements and tape measures. • Not adjustable. Each tape is locked in on the top, middle, and bottom • Using a finger monitor could be too big for an infant/toddler. Adding adjustability could alter accurate results

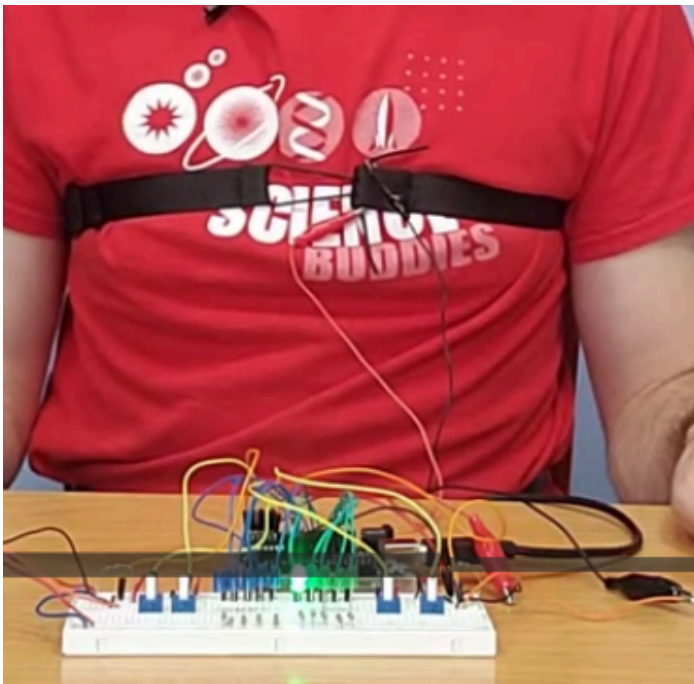
Design 5 - Overall Digital Tape Measure



This design utilizes arduino for the function for the measurement of the chest circumference and breathing rate. The digital arduino tape measurement is able to measure distance by the amount of rotation of the blue wheel, each time it spins more distance is covered, resulting in the tape measuring counting the distance traveled.



As the person breathes the string on the overall vest will contract and the digital tape measure will take a measure on how many inches the difference of when inhaling and exhaling.



This device incorporates a breathing monitor. The breathing monitor will be able to measure the breathing rate of diaphragmatic breathing. As the patient breathes the band will stretch the sensors connected to the band, and take readings of the resistance change.

This concept would be easy to remove with the clip on straps and easily adjustable to take reading on the upper or lower chest.

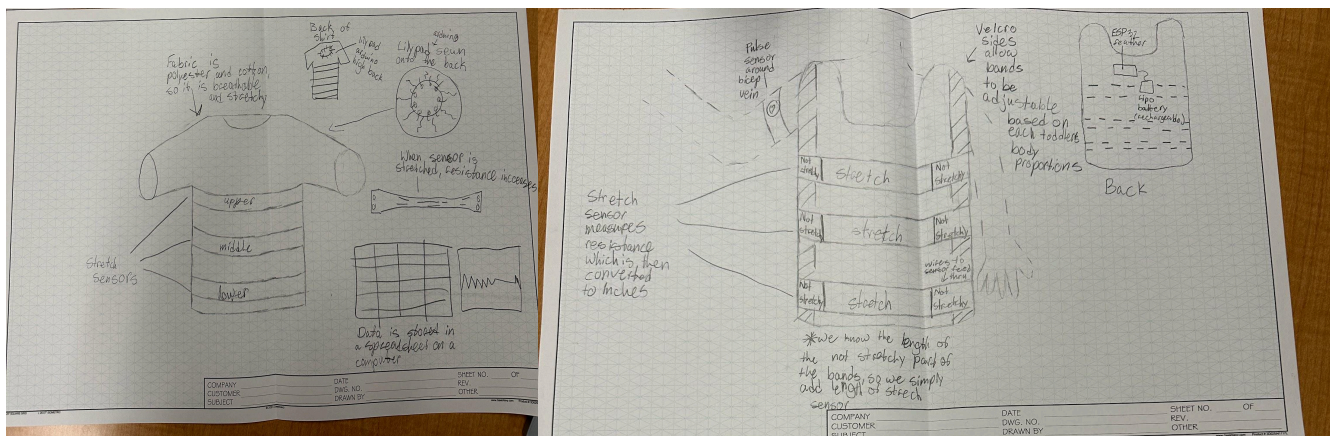
Some disadvantages of this device is that the arduino would be difficult to program and implement for this concept. This device may not be durable because it would be made out of wires and fabric and could be vulnerable to the outside environment. This device would not be

able to store data, the data would have to be recorded by hand by the measurements of the digital tape measurement.

Strengths	Weaknesses
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- The arduino of this device would take accurate measurements.
 - Easy to remove and adjust without much effort
 - Simple in design in that this design would just be an overall vest with a Stretch Sensor band.
 - This device can be used for multiple use in measure breathing rate and chest circumference
- The arduino would be difficult to program and implement for this concept.
 - This device may not be durable because it would be made out of wires and fabric and could be vulnerable to the outside environment.
 - This device would not be able to store data, the data would have to be recorded by hand by the measurements of the digital tape measurement.

Design 6 - Stretch Sensor Belt

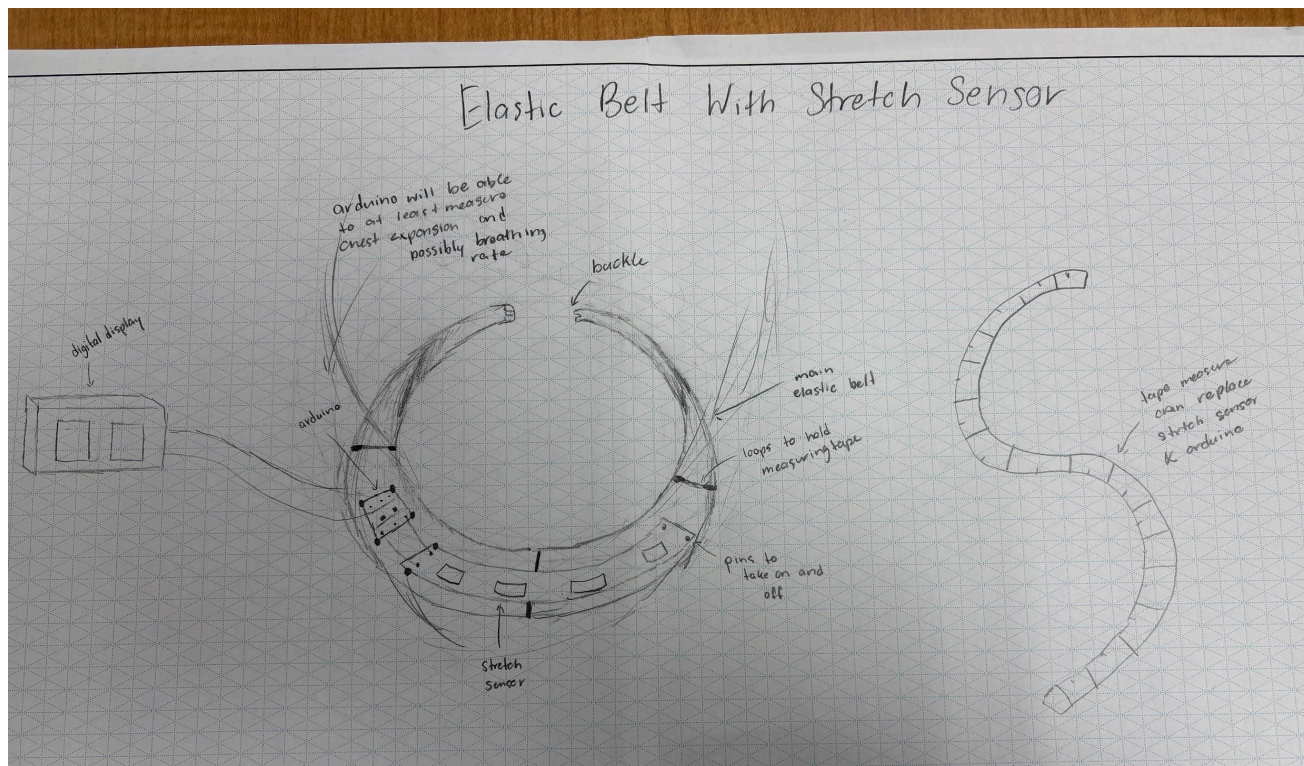


This design utilizes Stretch Sensors that change in resistance depending on the amount the elastic band is being stretched. This change in resistance will then be converted into a length measurement used to report the chest circumference of the user. This allows the ESP32 feather to record the starting and ending circumferences. The data can then be stored on the ESP32 feather then uploaded to a computer where it can be formatted within a spreadsheet. The power will be supplied by a lipo battery (rechargeable)

The device is a stretchy shirt made with a combination of polyester and cotton to ensure breathability, with 3 Stretch Sensor belts for the upper, middle, and lower chest portions. This allows for simultaneous data collection that can then be shown on a spreadsheet in their respective categories.

Strengths	Weaknesses
• Can measure both breathing rate and chest circumference • Can be adjusted to measure upper, middle, and lower chest calibrated to multiple body proportions • Utilizes the ESP32 which can transfer data to devices using bluetooth and/or wifi	• Complex design • The ratio between resistance and stretch length can be variable • Numerous wires must be used to hook up the Stretch Sensors to the ESP32 making the design pretty complex

Design 7 - Belt with Stretch Sensor/Measuring Tape

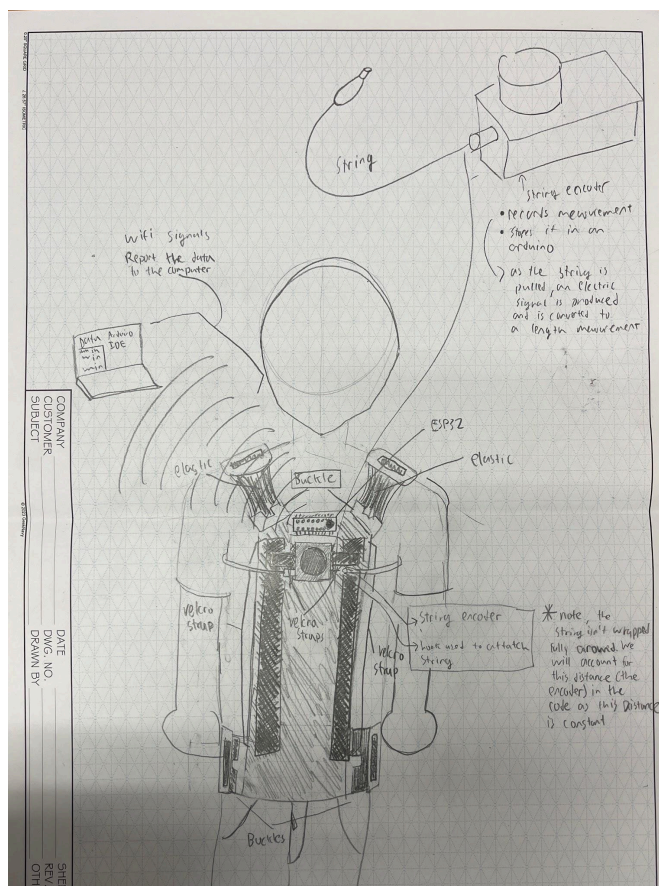


This design doesn't include a full piece of clothing with a system implemented; instead, it is just a belt that you would be able to put over a shirt or your body. This makes it more accessible and portable that would make it easier to take around. The fact that the entire design is elastic helps to fit around all body sizes without having to adjust it making it extremely convenient and practical. If needed to take manual or calibration measurements, there is a tape measure slot to slide through as well. It uses an elastic Stretch Sensor to measure resistance, which is passed

to the arduino to convert that resistance measurement to a length. That length is then added onto the set length of the elastic belt, excluding the length of the Stretch Sensor that is overlapped. That will give us an exhale and inhale length and then you can just subtract the 2 to find the difference between breaths. Additionally, the arduino can also measure heart rate if time allows for it and depending on our skills.

Strengths	Weaknesses
• Accurate measurements of additional features like heart rate without the use of a separate system due to the use of arduino. • Easy to wear with it being elastic and hold onto and wrap around any body size more easily along with the adjustable velcro or buckle strap in the back to tighten or loosen.	• It would be difficult to code and produce. Using the arduino and taking apart the parts of the device would be difficult to clean and could at risk of damage and messing up the system. • The external wired connection to the digital display could be inconvenient that can be exposed to the patent and outside environment, prone to damage.

Design 8- Draw Wire Encoder



The string in the encoder rotates the encoder as it is pulled. The rotation is converted into an electrical signal. BY recording the number of rotations, this can be converted into a length measurement

This design utilizes an ESP32 to store data using wifi. This eliminates the need for a direct connection to the computer or software device, making it easier to store the encoder on the vest. The encoder will be housed in the center of the vest and will be able to adjust up and down to measure different sections of the chest. Two long rectangular pieces of velcro will be attached to either side of the vest to allow for the encoder to be adjusted vertically.

Similar to the Tape Measure vest, the String encoder vest will be able to adjust to fit multiple body types and sizes.

Strengths	Weaknesses
- Accurate reading because of the direct relationship between the number of rotations and the length reported. - Easy to set up, the patient would just need to hook the string around their	bulky and could be difficult to put on because it could be heavy. The vest itself could hinder breathing if not fitted correctly on the patient or is too heavy; the straps exert a force on the chest that could prevent expansion.

torso to begin taking measurements. There is no additional complex set up. • Adjusted to different patents regardless of body type and size and can adjust measure to sections of the chest.	• Complicated to make and assemble to make a design to house the encoder. Also this device would be expensive to produce • Draw wire encoders are usually listed for at a minimum at 50\$ to 100\$ and more. • There could be uncertainty in measurements because the encoder would be supported by velcro, which can be subjected to ill alignment.
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D2. Design Solution Choice

[Decision Matrix:](#)

	Digital Tape Measure			Stretch Sensor		
QODs	Weight	Rating	Score	Weight	Rating	Score
Easy to Use	25%	3	0.75	25%	4	1
Adjustable (fitting)	12%	4	0.48	12%	4	0.48
Simplistic (Manufacturing)	8%	1	0.08	8%	3.2	2.4
Durable	25%	1	0.25	25%	3	0.75
Easy to Interpret Data	30%	2	0.6	30%	1	0.3
Total Score		TOTAL:	2.16		TOTAL:	4.93

Belt Design			String Encoder			Lily Pad Stretch Sensor		
Weight	Rating	Score	Weight	Rating	Score	Weight	Rating	Score
25%	3	0.75	25%	2	0.5	25%	3	0.75
12%	4	0.48	12%	3	0.36	12%	3	0.36
8%	1	0.08	8%	1	0.08	8%	1	0.08
25%	2	0.5	25%	2	0.5	25%	2	0.5
30%	4	1.2	30%	3	0.9	30%	3	0.9
	TOTAL:	3.01		TOTAL:	2.34		TOTAL:	2.59

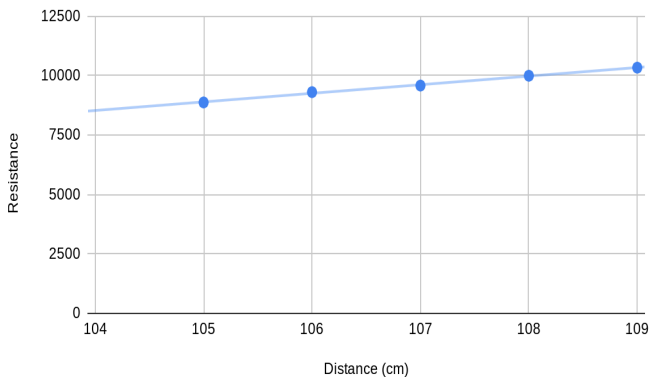
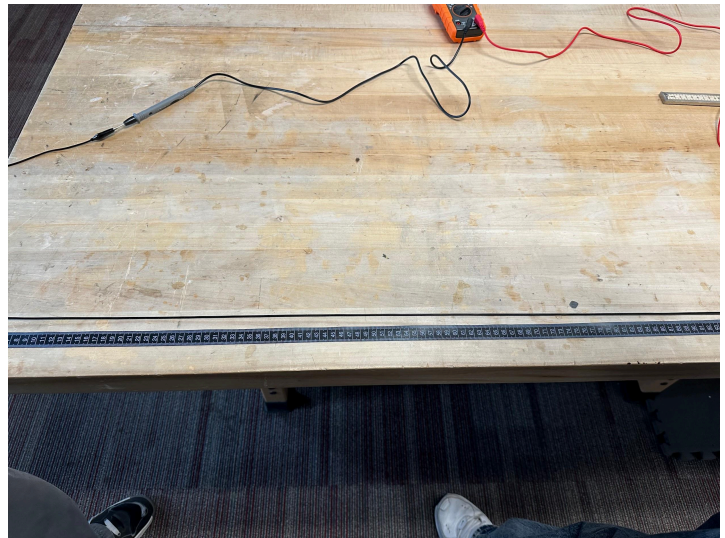
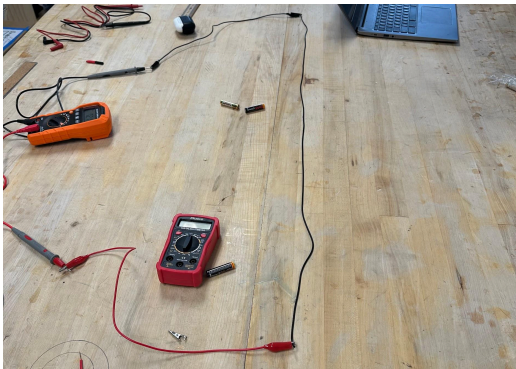
Digital Tape Measure	Elastic Belt	Draw Wire Encoder	Lily Pad - Powered Stretch Sensor	ESP32 - Powered Stretch Sensor
2.16	3.01	2.34	2.59	4.93

We ultimately went with the Stretch Sensor. Although the Stretch Sensor will require a great amount of calibration and data collection, we decided to choose it because the device can measure both the breathing rate and the chest circumference, without incorporating bulky components. For instance, the draw wire encoder is an exceptionally large piece of equipment, making it difficult to fit and adjust on the vest/onesie device. Among the designs of the decision matrix, the **Stretch Sensor (SS)** scored among the highest, which also aided in our choice. The Stretch Sensor also has the lowest cost out of all the designs. The conductive rubber band only cost \$9.95 and the belt straps and nylon fabric used will cost between \$10-\$20. The design is also highly simplistic as there are only two parts to the design: the strap buckle housing the Stretch Sensor, which will allow it to move up and down the nylon vest, and the vest itself. Based on the feedback from Dr. Broseman, the use of analog tape measures is strongly discredited. This means design 1, 3, and 4 cannot be used, as they all incorporate a tape measure as a means to measure chest circumference.

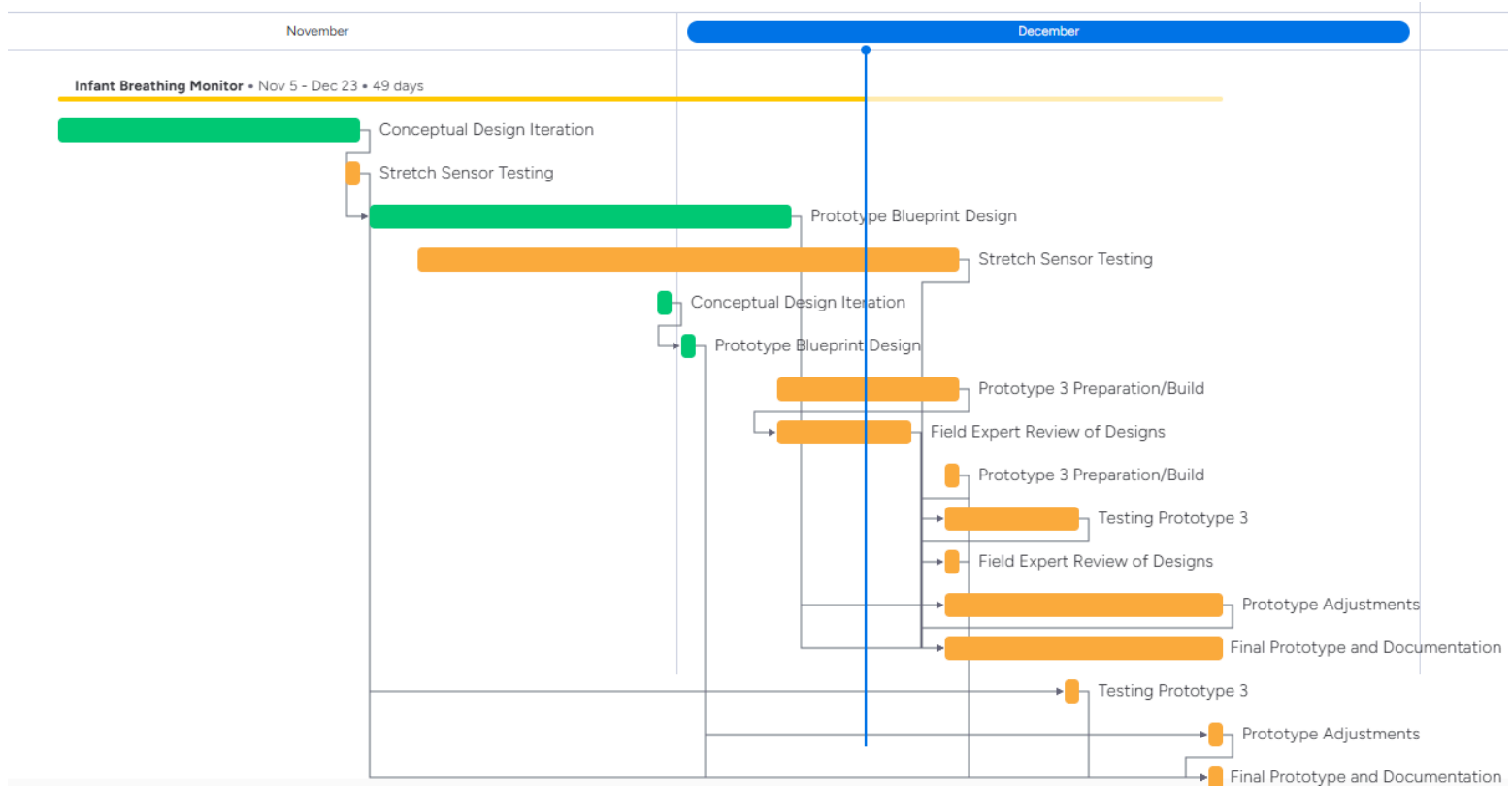
Plan of Action

We started our conceptual design iteration, ranking them based on our decision matrix and comparing based on our high priority design requirements. We determined that we will be using the Stretch Sensor design for our testable prototype; however, because there is no normative data on relating the stretch length of the stretch sensor to its resistance, we must create a preliminary prototype that we can collect data and establish our own relationship between the stretch length and the resistance via linear regression.

Creating a preliminary testable prototype is a critical step in determining the accuracy of our prototype. First we should collect data on the stretch length and resistance of the stretch sensor without the prototype. This includes securing one end of the stretch sensor band and stretching it to different lengths, recording the resistance at these lengths using a multimeter, then plotting the length vs. resistance points on a scatter plot:



Gantt Chart:



Stretch Sensor Testing Plan:

We must test the accuracy of the Stretch Sensor (ohms per inch) to see if it is applicable to our project.

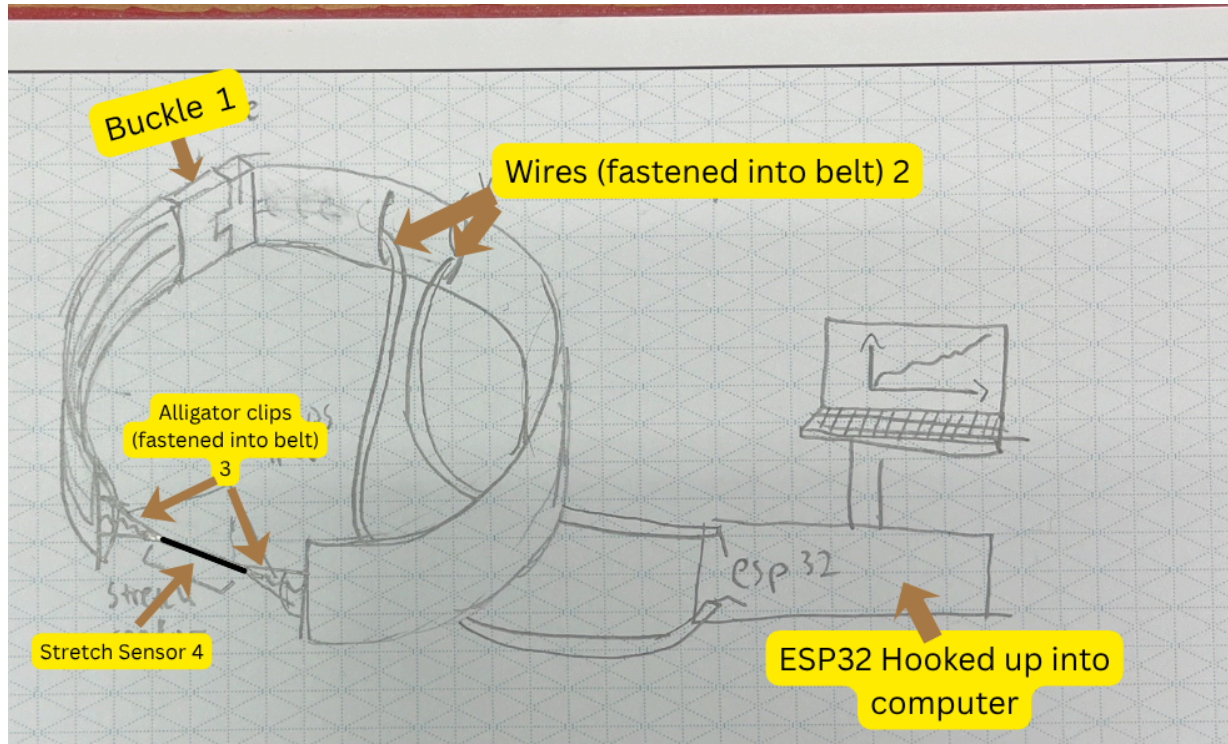
We will receive the sensor and do more research on the wiring and overall programming needed to get an output result in the arduino. We can then start doing more research into how sensitive and accurate the sensor is. Confirm that the sensor can measure the difference between its length at the beginning and middle of a stretch. To measure this stretch we could either make a breathing monitor prototype to conduct similar outcomes, or even connect different weights to the bottom to simulate stretch force and then support the weight.

If the Stretch Sensor proves to not be a viable solution for measuring circumference and difference in length during a stretch, then we could then reside to the draw wire encoder. While more expensive, it provides potentially more accurate results and adjustability. If/when we order the draw wire encoder and it arrives, it will require a similar testing and research process. Researching more into the wiring and connecting the sensor's reading into an actual tangible measurement. The measurement process will be more simple with this method by just pulling the wire to a specified length with a

ruler and then connecting that measurement to a formula that produces a measurement for the arduino. If we at least find some success in the Stretch Sensor where it can sense a stretch force, we then will incorporate this device into this design to measure/monitor breathing rate.

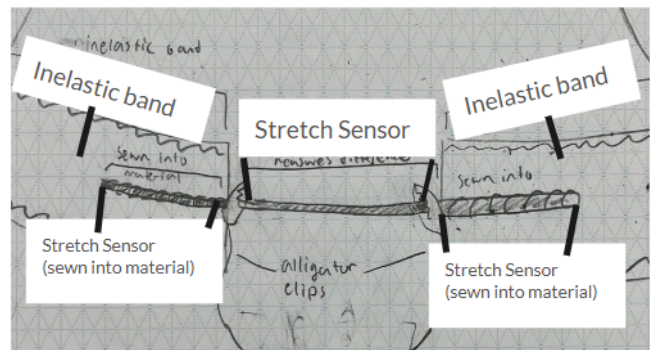
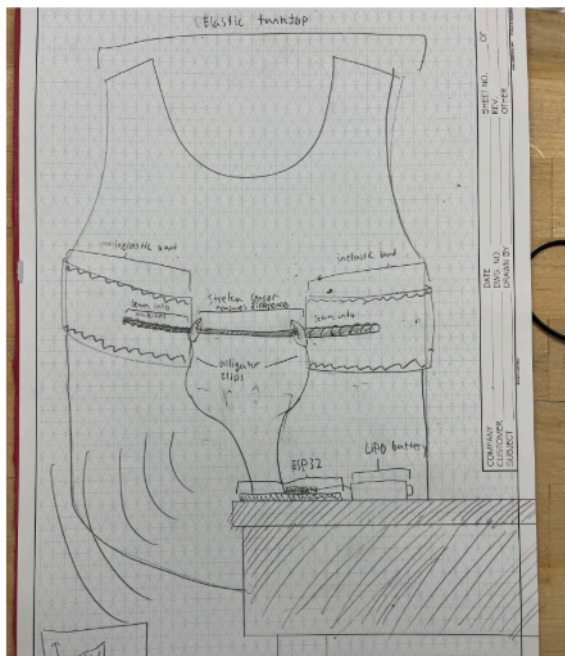
To optimize the accuracy of our design we are going to construct a prototype for our Stretch Sensor design. This design utilizes a buckle strap to house the Stretch Sensor, which wraps around the chest of the user. The sensor will be hooked up onto the ESP32 feather where our data will be stored onto the computer and we will be able to construct a calibration curve, relating the change in resistance of the band to the distance it stretches. This allows us to verify the accuracy and viability of the Stretch Sensor design and establishes a baseline for our data. The resistance in the Stretch Sensor is variable, thus a high amount of data is necessary to determine a trendline to relate resistance to distance stretched.

Prototype 1:



Prototype 2

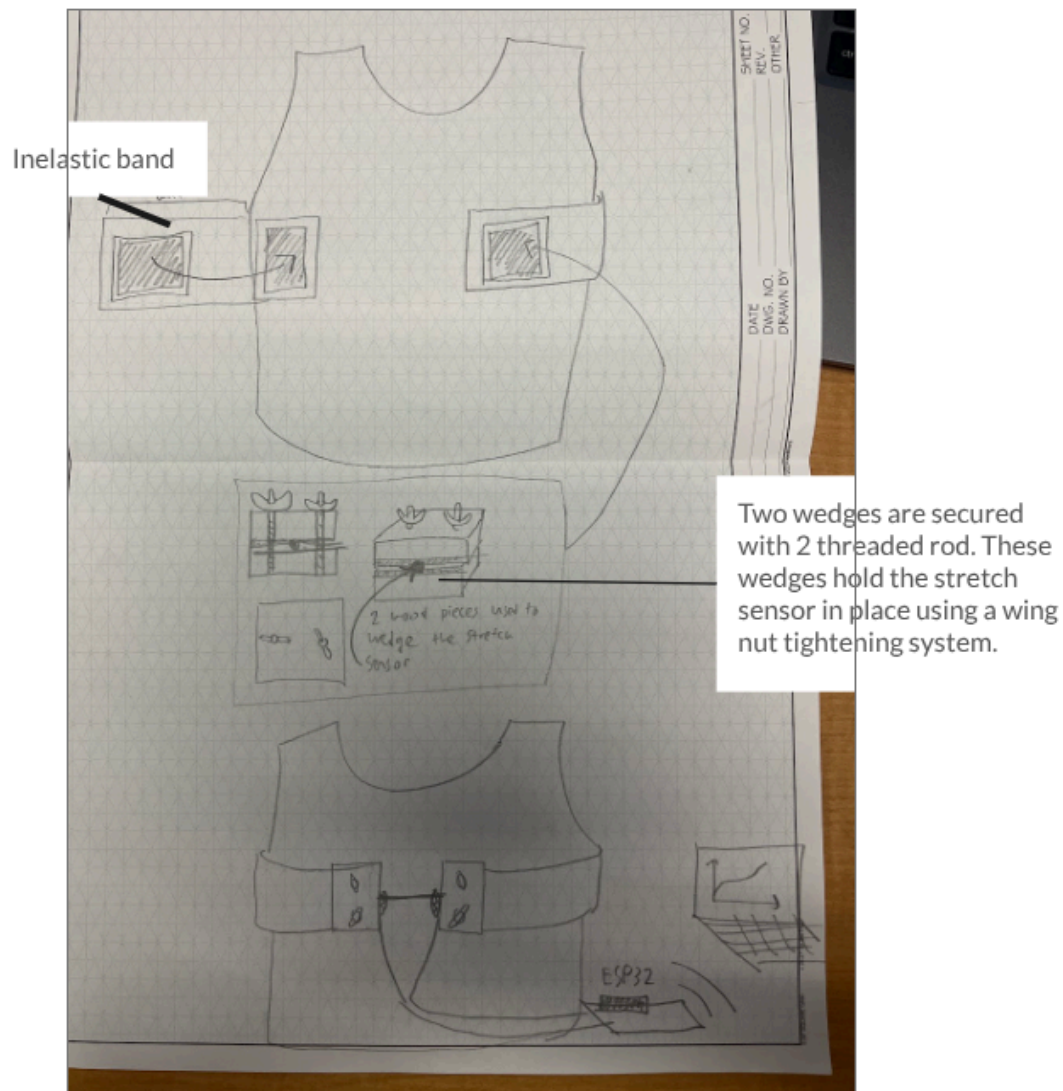
The tension in the Stretch Sensor cannot sufficiently provide enough support to keep the strap belt in place, thus a tank top will be used to hold the sensor in place instead:



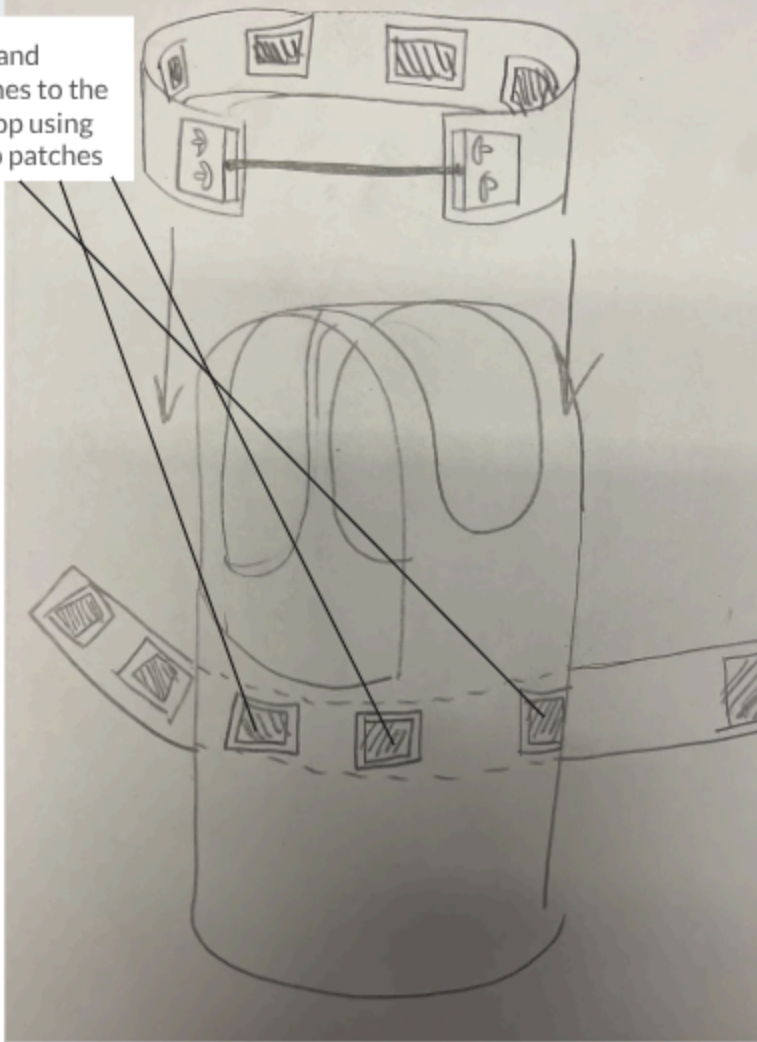
- Inelastic band allows for initial measurement.
- Stretch sensor measures the circumference accounted for stretch of patient breathing

Prototype 3 (present)

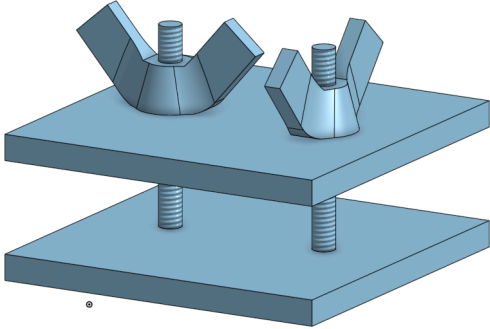
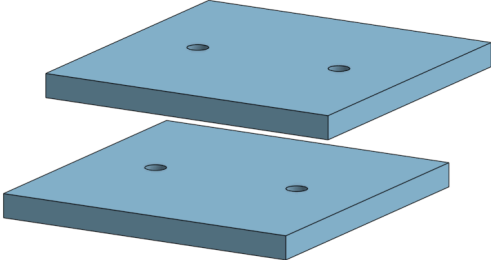
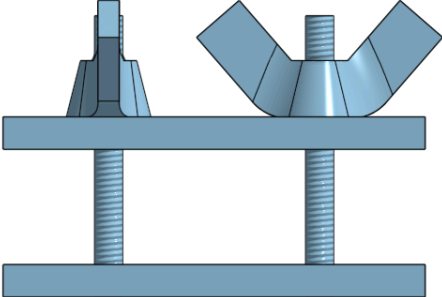
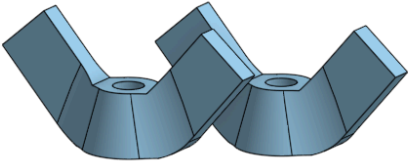
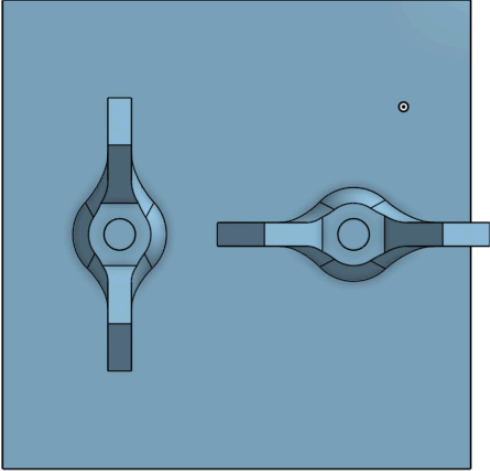
Directly sewing the inelastic band to the tank top could potentially restrict the user's movement and damage the Stretch Sensor when the user takes the tank top off: taking the tank top off with the Stretch Sensor still attached could lead to an overstretch. Prototype 3 aims at addressing this, adding velcro straps to attach the inelastic belt and to remove when finished.



The band
attaches to the
tanktop using
velcro patches



Onshape Drawing (Wedge)

Parts	Assembly
	
	
	

Assembled Wedge:

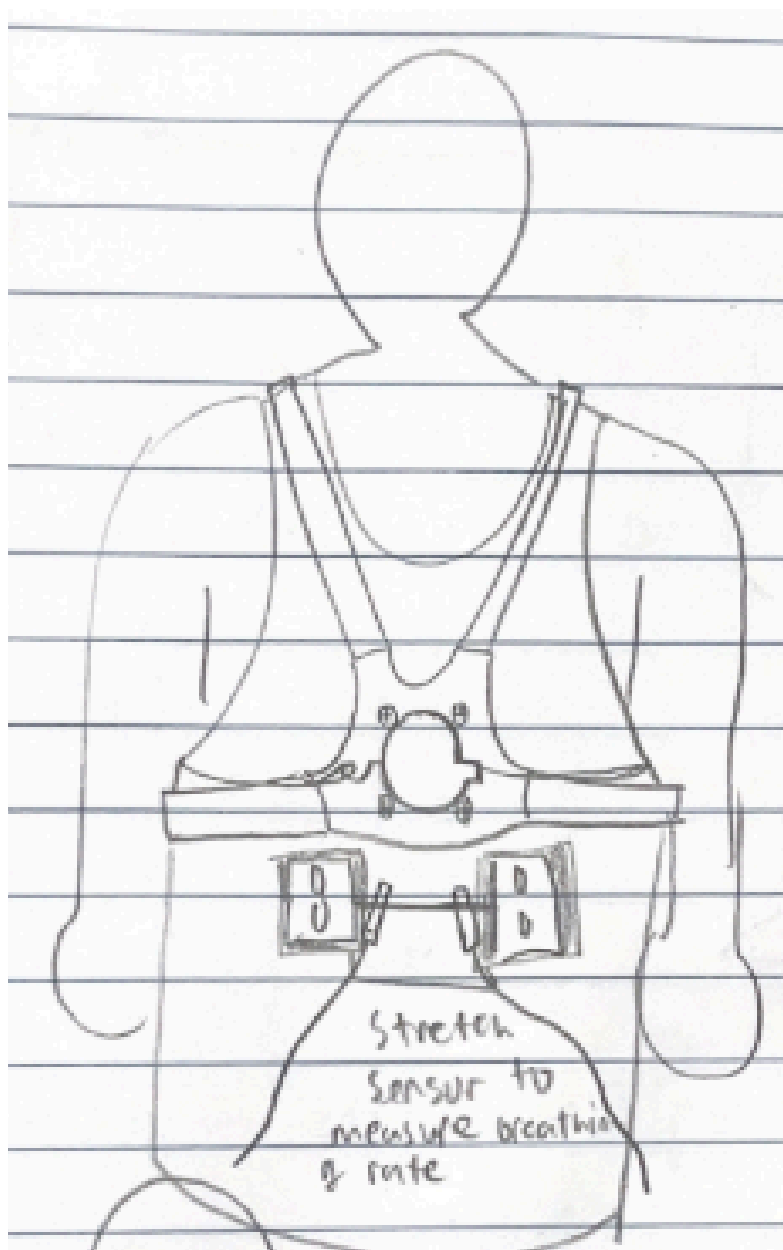


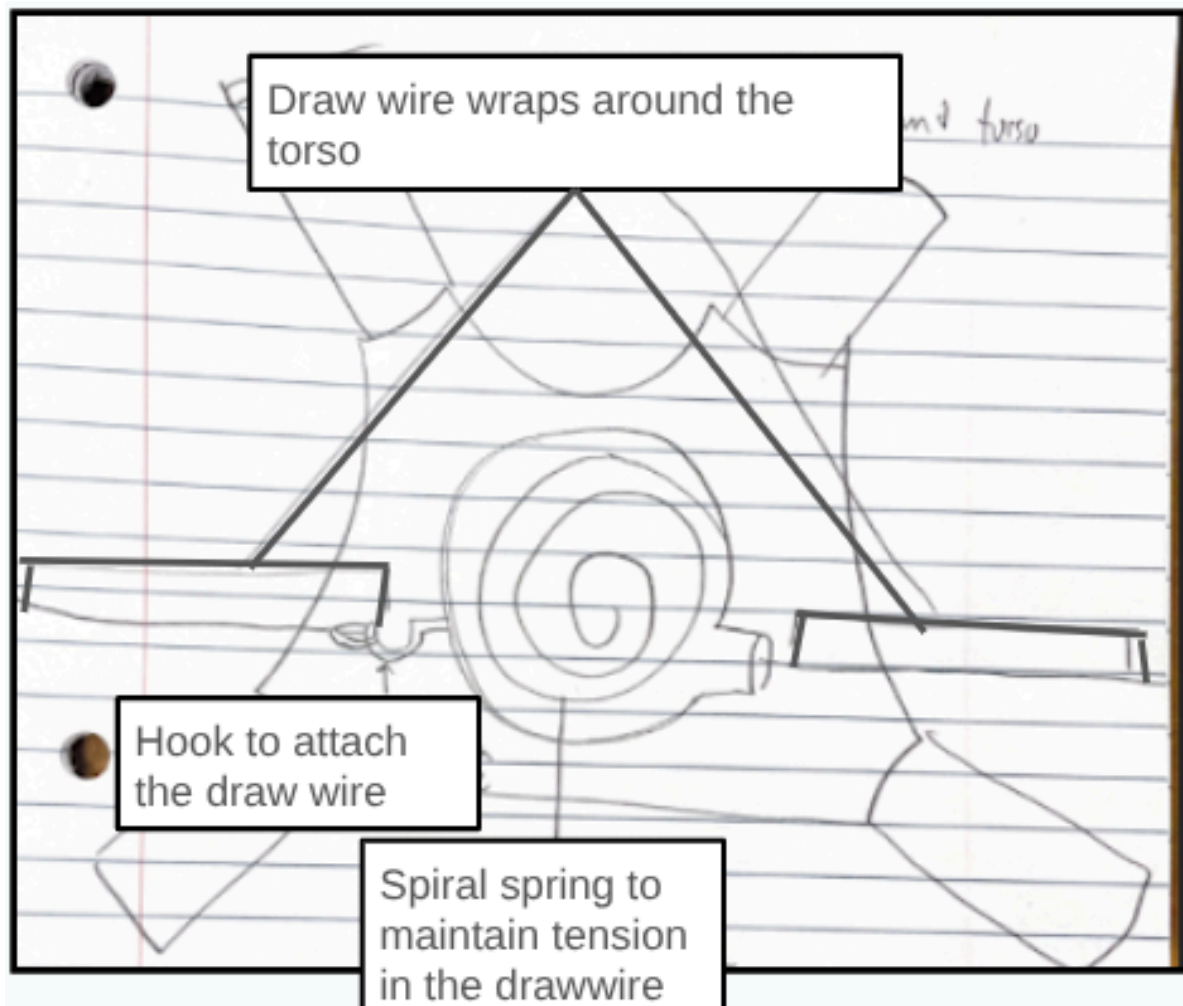
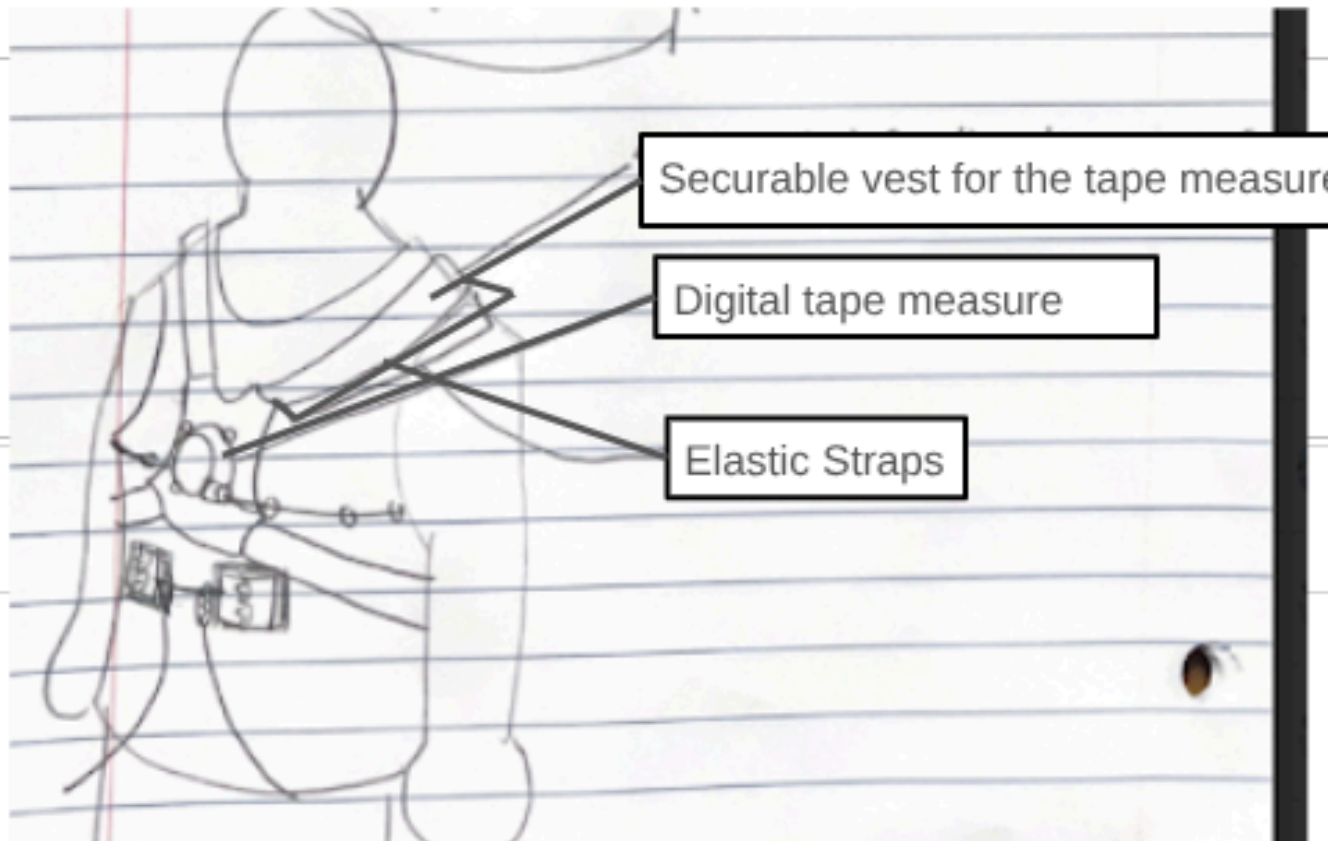
Semester 2

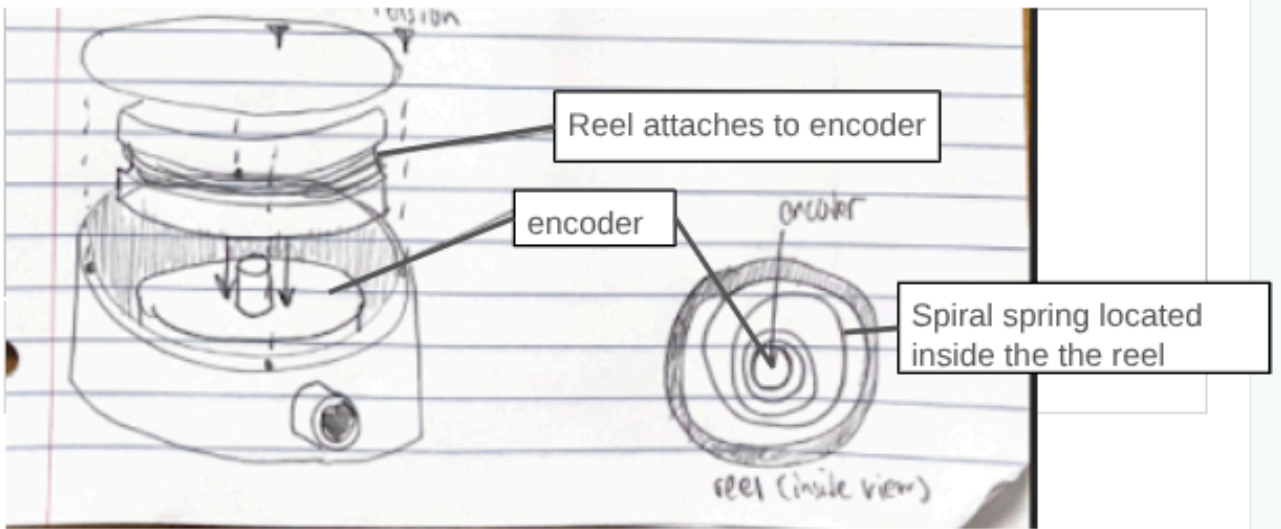
D1. Process For Generating & Comparing Possible Solution

Conceptual Designs

Draw Vest:

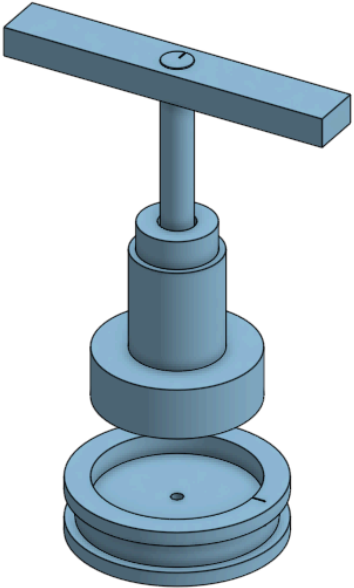






This design utilizes a digital encoder in order to measure the chest circumference. The encoder is located on the center of the user's chest and is fastened onto a vest with elastic straps. These straps can be adjusted in order to measure upper, middle and lower chest circumference. A stretch sensor will be attached to the tank top by using a similar design used in Prototype 3; however, it will only be used to measure the breathing rate.

Draw Wire Encoder Reel:

Parts	Assembly
	]

